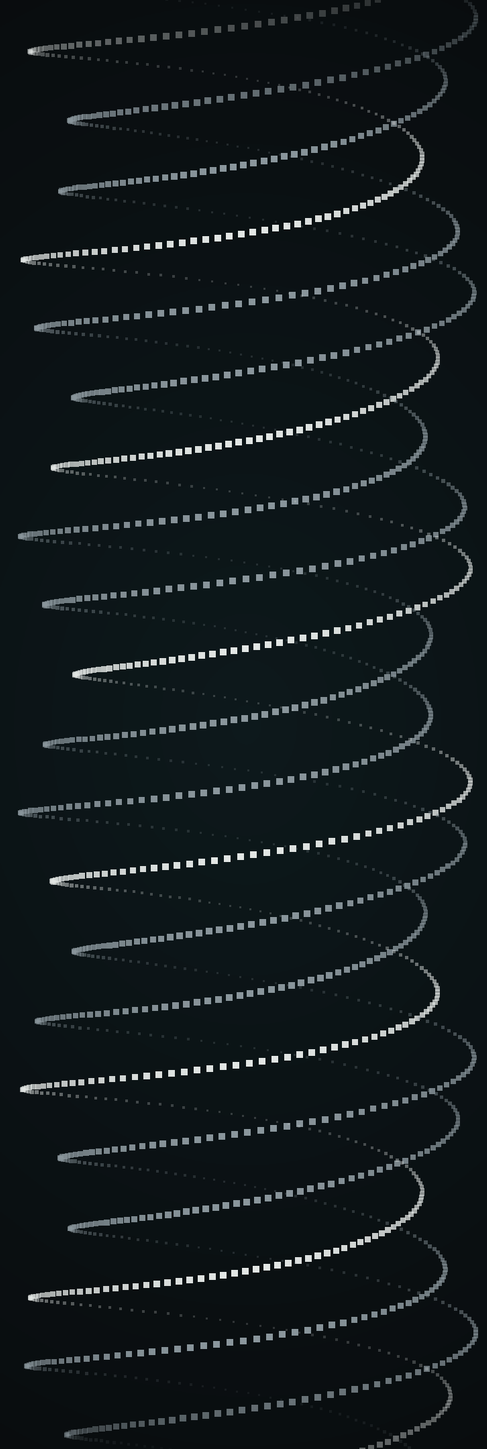
HALT

Security teams do not lack alerts.

They lack a defensible explanation of how those alerts connect.

Proof-Carrying Kill Chain Intelligence

CyberForge PS2: Kill Chain and Attack Path Analysis



THE PROMISE

Turn noisy logs into defensible attack stories.

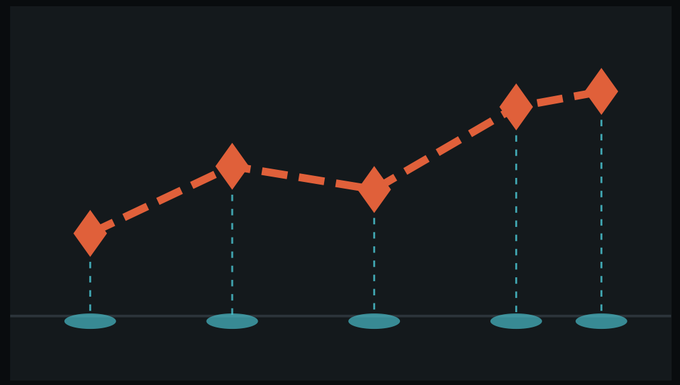
And refuse to invent one when the evidence cannot support it.

01	02	03	04	05	06	07
Add logs	Analyse	Investigate	SourceTrace	Verify	Response Handoff	Theatre

Three sources. Three honest answers.

10G

Reconstructed, unconfirmed incidents

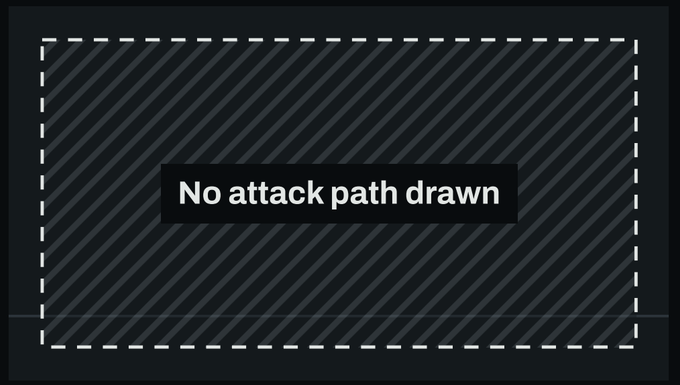


11 incidents

? UNCONFIRMED

30G

Reconstruction withheld

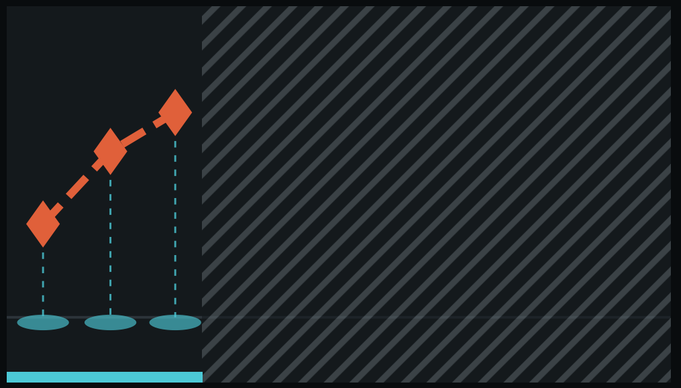


0 presented

▲ DEGRADED

50G

Partial reconstruction from a degraded source



11 incidents

▲ DEGRADED

Measured, not promised.

Plan	Parse	Prepare	Reconstruct	Explain
Whole-file SHA-256 and chunk boundaries	Workers write Parquet, and invalid data is quarantined	DuckDB finds duplicates and time faults, with a memory limit	Rules join candidates only through shared entities	Graphs, hypotheses, stories and sealed receipts

	10G	30G	50G
Source received	6.54 GB	19.62 GB	32.70 GB
Valid records	9,125,382	8,138,878	61,970,767
Exact duplicates found	812,235	1,310,857	7,645,803
Duration	497.45 s	543.53 s	4,216.33 s
Peak memory, upper bound	3.62 GB	2.56 GB	6.44 GB
Incident open, from receipt	87.3 ms	Not applicable	79.52 ms

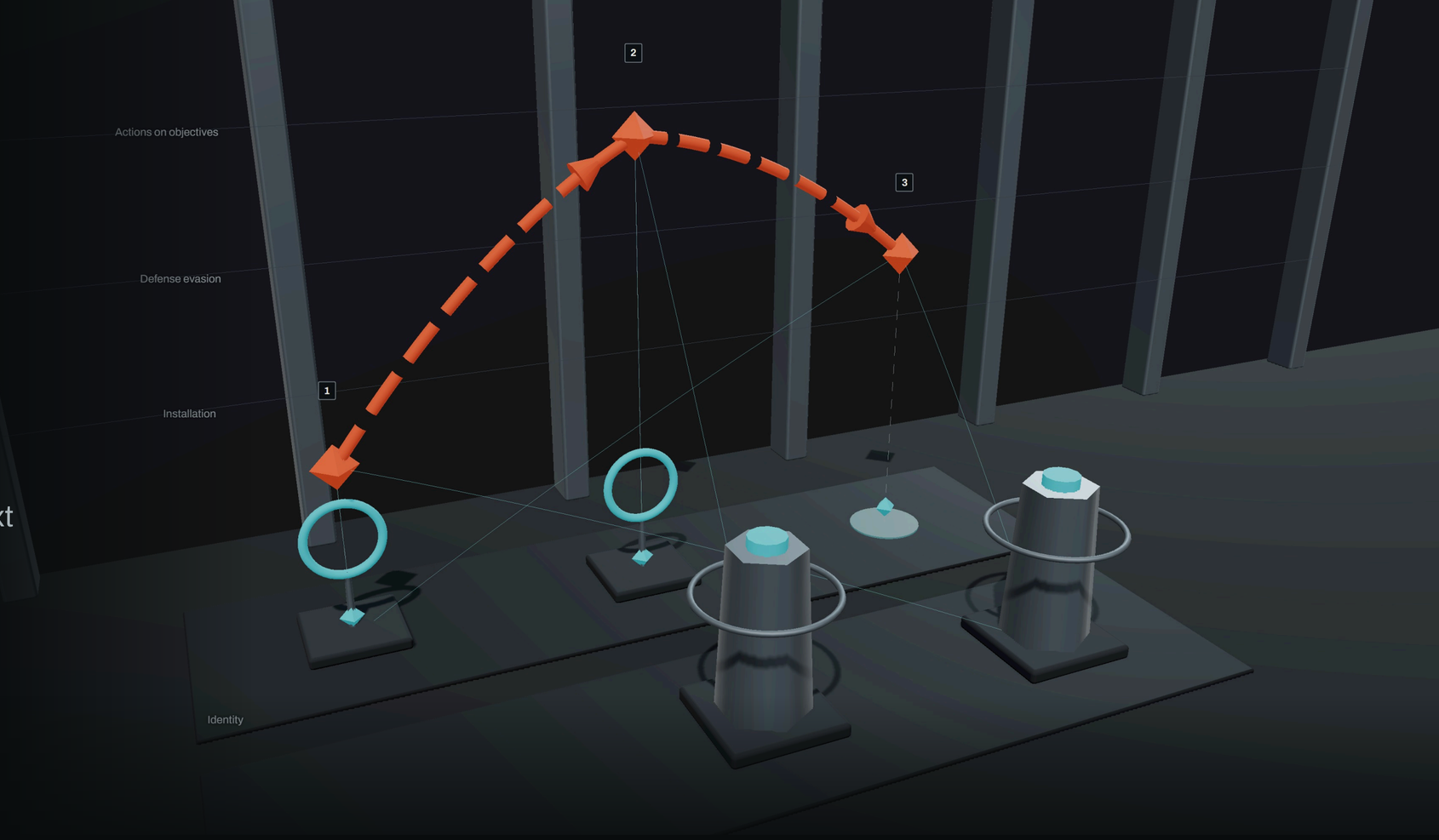
Recorded FireX results on the recorded machine. Not a performance guarantee.

SEMANTIC 3D ATTACK WORLD

Step inside the incident.

Every position means something.
Nothing is decoration.

Rendered from the stored 10G incident on the next slide.



Left to right	Height	Depth	
Source order, not a clock	Kill-chain stage	Organisation layer	**Ember** selected path **Cyan** observed **Violet** inferred **Gold** hypothetical

SOURCETRACE, ON THE STORED 10G RUN, TIMES IN UTC

One search. Back to the exact line.

`svc_backup30` — **An account, not a file**

RAW RECORD · LINE 3,226,313 · HASH MATCHES

```
date=2026-10-29 time=03:55:43 … logdesc="Object attribute configured"
user="Rakes" … srcip=203.228.177.117 … dstip=10.10.99.1 … action="Add" …
msg="Add system.admin svc_backup30"
```

03:55:43	**Account added** Observed, line 3,226,313	
	↓ Connection inferred by a rule	
04:02:01	**The new account signs in** Observed, line 3,227,163	
	↓ Connection inferred by a rule	
04:07:25	**Account deleted** Observed, line 3,227,930	

1. Source record
2. Parser mapping
3. Canonical event
4. Duplicate and burst handling
5. Detection rule
6. Evidence relationship
7. Hypothesis
8. Incident statement
9. Proof receipt

606 of 606

cited records resolve to their source line, in 10G and again in 50G

The events exist. Malicious intent is not established.

It knows when to stop.

30G: Reconstruction withheld because the source does not provide a trustworthy time axis.



21.89% valid log records

8,138,878 records at one instant, `2026-10-30T08:48:36Z`

78.11% quarantined

Zero incidents does not mean zero threats.

50G: Partial. Every incident rests on the part with usable time.



29.51% usable time order, 18,289,294 records

70.49% collapsed time, 43,681,473 records

Activity among the collapsed records is unknown, not absent.

Verified file. Unconfirmed conclusion.

TRUST CENTRE

Source integrity	✓ VERIFIED
Processing integrity	✓ VERIFIED
Reconstruction trust	! NEEDS REVIEW
Ground-truth validation	– NOT MEASURED



Proof receipts valid in both 10G and 50G.
A seal shows that a receipt has not changed. It does not show that a conclusion is true.

RESPONSE HANDOFF

■ NOT EXECUTED

Investigation actions
From cited records and rival hypotheses

Missing evidence to collect
Gaps recorded in the incident

Identities and assets to validate
Named by the evidence

Reachable systems to monitor
Hypothetical, and labelled so

Approval requirements
No language model chooses a response

Ready for the next log. Honest about the edge.

Live Intake

01	02	03
Add data	**Start analysis**	**Open results**

Allowlisted folder, or resumable upload

Source hash, format and sample records shown first

Each source gets its own isolated run

Recovers from interruption, on the same job system

Analyses FortiGate key-value logs. CSV and JSON are detected and refused with a clear reason.

What we do not claim

Detection accuracy is not measured: no organiser ground truth was provided.

Every incident stays unconfirmed until an analyst reviews it.

These logs have no payload or endpoint data, so they cannot prove an action succeeded.

50G took about 70 minutes on the recorded machine, over the one-hour target.

Peak memory is an upper bound, not a sampled total.

The synthetic response lab is a separate demonstration, not derived from these logs.

Contain the threat. Continue the business.

From raw logs to evidence-backed attack paths.

